

Address Spaces

Definition

An **address space** is the set of memory addresses a process is allowed to use. It is one of the most important concepts in memory management of an operating system.

1. Basic Idea

Main Concept

Every computer has **main memory** for holding executing programs. In a very simple operating system, only **one program** can stay in memory at a time. If another program has to run, the first one must be removed.

Need for Protection

Modern operating systems allow **multiple programs** to stay in memory at the same time. To prevent them from interfering with each other and with the operating system, a **protection mechanism** is required.

2. Memory Protection

Protection Mechanism

The protection mechanism is implemented in **hardware**, but it is **controlled by the operating system**. This ensures that one process cannot wrongly access the memory of another process.

Why Protection is Needed

Without protection:

- one program may overwrite another program,
- one process may damage the operating system,
- the system may become unstable or crash.

3. Address Space of a Process

Process View

Normally, each process has its own range of addresses, usually starting from **0** up to some **maximum value**. The process can use only addresses inside this range.

Simple Case

In the simplest case:

- the process address space is smaller than main memory,
- the whole process can fit in physical memory,
- execution becomes straightforward.

4. Problem of Large Address Space

Modern Computers

Modern computers commonly use **32-bit** or **64-bit** addresses.

$$2^{32} \text{ bytes} \quad \text{or} \quad 2^{64} \text{ bytes}$$

This gives a very large possible address space.

The Problem

Sometimes a process may have an address space larger than the available main memory. Then the whole process cannot be kept in RAM at the same time.

Old Situation

In early computers, if a process needed more address space than physical memory, it simply could not run properly.

5. Virtual Memory

Definition

Virtual memory is a technique in which the operating system keeps part of a process's address space in **main memory** and part on **disk**, and moves pieces back and forth when needed.

Main Idea of Virtual Memory

Virtual memory creates the **abstraction** that a process has a large continuous address space, even if all of it is not physically present in RAM at one time.

How It Works

The operating system:

- keeps some parts of the address space in RAM,
- keeps the remaining parts on disk,
- transfers data between RAM and disk as required.

6. Address Space vs Physical Memory

Important Difference

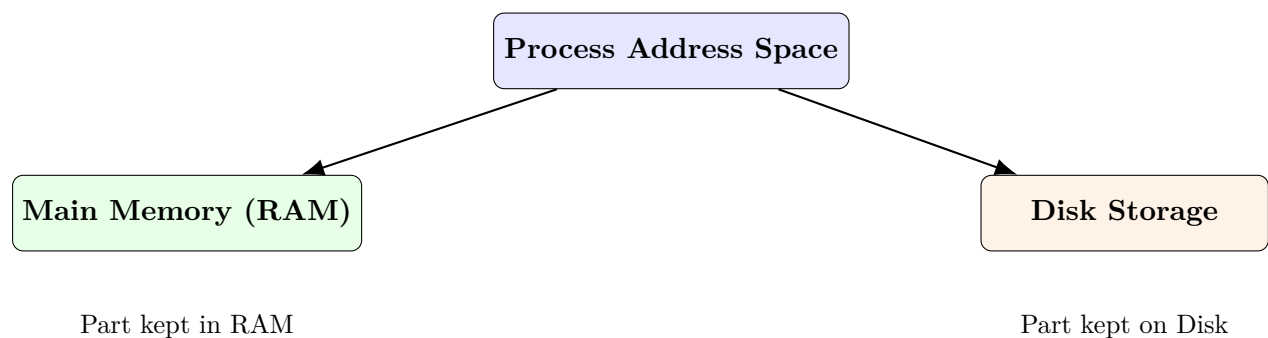
The **address space** of a process is a logical view, while **physical memory** is the actual RAM installed in the machine.

Key Observation

The address space is **decoupled** from physical memory. Therefore, it may be:

- **larger** than physical memory, or
- **smaller** than physical memory.

6.1 Simple Diagram



7. Role of the Operating System

OS Responsibility

Managing **address spaces** and **physical memory** is a major responsibility of the operating system.

Main Tasks of OS

The operating system must:

- manage memory allocation,
- protect processes from one another,
- map address space to physical memory,
- move data between RAM and disk when necessary.